Research title

UNderstanding and **I**mproving communication about blood **TE**sts in primary care **(UNITE)**: a multi-method study using ethnographic and participatory approaches

Research career to date (1000/1000 words)

Research you have led or made a significant contribution to (making clear what your role was), the research methods you have experience of, and the impact and outputs of the research you have been involved in (linking to details given in your CV). Any other relevant experience such as supervisory, management, teaching, public engagement and involvement, or leadership experience. Other skills and experience which highlights your suitability for the fellowship and demonstrates your potential as a future health or social care research leader

Academic Clinical Fellowship, KCL (2021-2025)

I have been a Research Fellow since August 2023 on UKRI-funded Social Responses to Stigma project, led by Dr Andrew Guise. This ethnographic study examines how stigma and discrimination arise within the health and care services used by people who are homeless in south London. My work includes ethnographic observation in day centres and collaborative analysis of fieldnotes and interviews.

Outputs:

- I lead a research strand exploring the intersection of stigma, homelessness and health self-management. I presented this work at the RCGP Conference 2024 and am preparing a first-author paper for submission to *Sociology of Health and Illness.*
- I co-authored a paper on a nested quantitative survey of homeless people's stigma experiences, under review at *BMJ Open*.
- I am the co-author of a forthcoming submission on drug use stigma *(International Journal of Drug Policy)*.

In 2022, I collaborated with early-career researchers at Charité University Berlin and the Medical University of Vienna on a European University Hospitals Alliance project to improve Type 2 diabetes prevention in women with gestational diabetes. We secured a £10,000 seed grant and convened workshops in London, Berlin and Vienna. I am the first author of our interdisciplinary commentary paper, currently under review at *Journal of Midwifery and Women's Health*.

Other activities:

- I am studying for a PGCert in Applied Research Methods at KCL, due for completion June 2025. I have been awarded Distinction in all completed modules.
- I am an advisory board member (2022-) for PREPARED, a multi-method study of a primary care intervention to improve pregnancy outcomes in women with diabetes.
- I was invited by Professor David Misselbrook to be a peer editor for his textbook, *Thinking About Medicine: An Introduction to the Philosophy of Medicine*, providing a critical review of each chapter.
- I am a peer reviewer for publications including *Social Science and Medicine, British Journal of General Practice, Journal of Medical Ethics, BMC Medical Ethics, Digital Health,* and *Information, Communication & Society.*

MPhil, University of Cambridge (2019-2020)

My interdisciplinary MPhil (First Class with Distinction) combined medical philosophy, sociology and anthropology. I received a Starred Distinction for my thesis, a qualitative interview study exploring how participating in an online gestational diabetes support community shaped women's experiences of

pregnancy. I published this work as a single-author publication in Sociology of Health and Illness (2022) and presented it at three international conferences, including FIGO World Congress (2021).

Overdiagnosis research (2019-)

During the MPhil I became interested in how clinicians negotiate risk and uncertainty and communicate these issues to patients. I published my research on how to explain the pitfalls of screening to patients in *Journal of Medical Ethics* as a single-author paper, which scored in the top 5% of research outputs on Altmetric. I was invited to present this work at Preventing Overdiagnosis 2022.

I then completed a Diploma in the Philosophy of Medicine, exploring in my dissertation how clinicians balance risk in end-of-life care decisions. I was awarded the Sydney Selwyn prize for best Diploma performance and invited to give a public lecture on my dissertation research. I published this work as single-author in *Journal of Evaluation in Clinical Practice* (2022) and presented at Preventing Overdiagnosis 2023. I belong to the Preventing Overdiagnosis Early Career Researchers group and am organising a pre-conference workshop for the 2025 Oxford conference.

Previous research experience

As an Honorary Research Fellow at QMUL (2017-2019), I led a systematic review of post-partum depression and intimate partner violence in high-income countries, winning 1st prize at the British Society for Biopsychosocial Obstetrics and Gynaecology conference (2018). I co-authored a systematic review and meta-analysis of health service use by people with severe mental illness, published in *PLOS Medicine* (2021). I also led the quantitative analysis of the LGBT Foundation's survey of women who have sex with women (2019), the results of which were presented at the All-Party Parliamentary Group on Sexual and Reproductive Health.

For my MSc dissertation I conducted a quantitative analysis of the National Survey of Sexual Attitudes and Lifestyles, focusing on birth order and sexual health outcomes. This work – published as a first-author paper in *Sex Education* (2018) – attracted significant media attention, including coverage in the Independent and the Daily Mail. (https://www.independent.co.uk/news/education/education-news/sex-education-first-born-children-parents-families-london-school-hygiene-topical-medicine-a8558211.html).

At medical school I co-led a national survey of medical students' views on abortion teaching, which I presented at two international conferences (2016). I was subsequently invited to the Royal College of Obstetricians and Gynaecologists to discuss curriculum recommendations. I also led a hospital-based audit of the risk assessment of venous thrombo-embolism (VTE) in pregnancy which led to the hospital updating its VTE guidelines; I presented this work at two national conferences (2015).

Teaching and management

In 2018, I co-designed and delivered a critical appraisal module (11 weeks) for Global Health BSc students at QMUL. I was awarded the Foundation School Award for Teaching Excellence in recognition of my mentorship of medical students applying for Academic Foundation posts. I have held Associate Fellowship of the Higher Education Academy since 2019.

As chair of the Cambridge Global Health Lecture committee (2019-2020), I coordinated a series of six public lectures, attracting over 200 attendees. From 2015-2016, I was Policy and Advocacy Director for Students for Global Health UK, leading a team of policy analysts and steering four national student research collaborations.

Impact of fellowship

An NIHR Doctoral Research Fellowship will allow me to continue my academic career and develop as a future research leader. My eventual goal is to become a Professor of Primary Care, using interdisciplinary qualitative methods to explore what good primary care looks like in practice. The

proposed project will develop my skills in complex qualitative methodologies and allow me to build my portfolio of research addressing medical overuse. Following the PhD, I hope to secure post-doctoral funding (e.g. an Academic Clinical Lectureship) to conduct further ethnographic work exploring the social context of medical overuse. I am especially interested in co-producing creative outputs which can engage wider audiences to consider the wise use of medical resources.

Plain English summary (628/600 words)

Aims, background, design and methods used, PPI involvement, dissemination.

Background
The number of blood tests ordered in general practices has tripled since 2000. Blood tests can be useful but do not always make patient care better. They can sometimes cause problems: for example, borderline blood tests may mean that patients worry or have unnecessary further tests. General practice staff don't always talk clearly about these issues, so patients may have different expectations of blood tests.

Research suggests that communication about blood tests in general practice could be improved. Patients say they sometimes feel uncertain about why tests are done and what results mean. General practitioners (GPs) often make decisions to do blood tests without much input from patients, who feel they don't get enough chance to discuss blood tests and results.

Improving communication between general practice staff and patients about blood tests could reduce unnecessary testing and help patients make more informed decisions.

Aims
This project will help general practice staff and patients talk more clearly about blood tests, so patients are more involved in deciding whether to have tests and can understand their results better.

Research design
This project has two parts:

1. <u>Understanding how blood tests are discussed:</u>
    i. *Evidence search:* I will look at existing research on how blood tests are discussed in general practices to understand what we know so far and what is missing.

    ii. *Observation at GP practices:* I will spend several months watching conversations about blood tests at two different general practices. This will include patient consultations. I will make recordings with the permission of staff and patients.

2. <u>Improving how blood tests are discussed:</u>
    i. *Workshops with general practice staff:* I show short clips of the recordings to practice staff. They will discuss their worries about how blood tests are discussed and think about how to improve communication.

    ii. *Forum Theatre workshops with the public:* I will work with a theatre group and public contributors to create a short play based on the findings of the project. The play will be performed to members of the public as part of a "Forum Theatre" workshop, in which attendees are invited to make changes to the play to improve how blood tests are discussed. With the support of public contributors, the play will be developed into two tools:
        - a workshop to help general practice staff speak more clearly about blood tests
        - an online resource for patients to help patients feel more confident discussing blood tests and understanding their results.

Patient and public involvement (PPI)
People who have recently had blood tests helped design this research and will stay involved throughout. This means the research can respond to the needs and preferences of people having GP

blood tests. To develop this proposal, I held three workshops with eighteen members of the public. They described many issues with communication about blood tests and felt strongly that this research was important.

Members of the public will be part of a PPI panel, meeting three times a year to guide the project by:

- reviewing consent forms/study information leaflets;
- helping to analyse data;
- writing summaries of the research for the public.

They will be paid for their contributions and will receive funding for training. They will also have the chance to help share the research through conference presentations. Two lay people will co-chair a separate Project Advisory Group to oversee the project.

Sharing the research

The research will be shared with local community groups, GP practice staff and patients via social media, blogs, a podcast, and a public event. The workshop tool created from the play will be used to train GP staff and trainees on better blood test communication. I will work with the PPI panel to develop leaflets and social media resources about the research, which will be made available online along with the play.

Scientific abstract (500/500 words)

Research question, background, aims & objectives, methods, timelines for delivery, anticipated impact and dissemination.

Background
Primary care clinicians order nearly half of all NHS blood tests, with primary care testing rates tripling between 2000-2016. Blood tests place significant demand on GP time and budgets but do not always improve patient care. Research suggests that communication about blood tests in primary care is often inadequate. The limitations of blood tests are seldom explained to patients, who report that they are not always involved in blood testing decisions and often do not understand what their results mean. With patients now able to access blood test results online, these communication issues need urgent attention. This research will use ethnographic methods to explore the role of communication in shaping blood testing practices and improve how blood tests are discussed in primary care settings.

Aim
To understand how GPs and other primary care clinicians discuss blood tests with patients and how this communication might be improved.

Research questions

a. How is the issue of blood testing communication framed in the existing literature?
b. What do clinicians and patients seek to accomplish with blood tests and how is this communicated?
    a. How are decisions about ordering blood tests negotiated and communicated in practice?
    b. How do clinicians and patients communicate what they understand about blood test results?
c. What concerns do primary care clinicians have about blood testing communication, and how can insights from participatory ethnography support clinicians to identify solutions to these concerns?
d. How can participatory creative methods be harnessed to support more effective communication between patients and primary care clinicians about blood tests?

Methods
<u>Phase 1: Understanding blood test communication</u>

- WP1: Meta-ethnography synthesising existing qualitative literature on blood testing communication.
- WP2: Detailed ethnographic observation in two GP practices with contrasting patient demographics, observing the entire blood testing process with a focus on patient consultations where blood tests are discussed.

<u>Phase 2: Improving blood test communication</u>

- WP3: Video-reflexive ethnography workshops with GP staff to identify concerns and challenges around blood testing communication as well as opportunities for change.
- WP4: Video-recorded Forum Theatre workshops with members of the public and GPs. The recorded performances will be adapted into a workshop resource (for clinicians) and an online resource (for patients/the public) to support improved blood test communication in primary care.

Timelines

- Year 1: meta-ethnography, ethics application
- Year 2: detailed ethnographic observation and video-reflexive ethnography workshops
- Year 3: creative co-production of patient and clinician resources

Anticipated impact
This participatory research project will reframe the GP consultation as a key site of opportunity for the prudent use of blood testing. It will engage GP practice staff, lay people and theatre-makers to co-produce resources to support patients and primary care clinicians to communicate better about blood tests.

Dissemination

The workshop resource developed in WP4 will be shared with local primary care staff and GP trainees, as well as at national conferences. A version of the workshop will be made available as a publicly-accessible resource for patients. Research findings will be published in academic journals and shared via social media, blogs and podcasts.

Detailed research plan (5250/5000 words)

1. What is the problem?

Nearly half of all NHS blood tests are carried out in primary care, and primary care test requests tripled between 2000-2016.[1] Primary care blood testing is expensive[1] and managing test results contributes significantly to GP workload.[2] Despite their financial and administrative burden, blood tests do not always improve patient care: many blood tests do not change clinical management, and up to 30% of blood tests may be unnecessary.[3] Blood tests, like any medical intervention, can also cause harm. False-positive or borderline results may cause unnecessary anxiety or lead to further unwarranted procedures, while normal results may provide false reassurance which can delay diagnosis.

It is important that primary care staff involve patients in decisions about blood testing and explain clearly what test results mean. Yet many patients say they struggle to understand their results and do not get a choice about whether or not to do blood tests.[4] While GPs acknowledge that tests are not always useful and can sometimes lead to more uncertainty,[5] they often initiate blood tests without discussing alternatives or explaining the potential risks and limitations.[6] This limits opportunities for patients to be involved in their healthcare and make the right decisions for them.

NHS England has produced guidelines for optimising blood testing in primary care, but these focus on clinical appropriateness of blood tests; [7] the guidelines do not address *how* communication about blood testing should proceed, nor how to explain the risks and pitfalls of testing to patients. Recommendations about which tests to perform are useful, but cannot replace the need for GP staff to elicit patients' expectations of blood tests and explain the reasoning behind testing decisions.

Examining what happens in GP consultations about blood tests is crucial to understanding the rising demand for blood testing in primary care. However, research which directly explores these interactions is limited, meaning that there is a gap in our understanding of how communication about blood testing might be improved. This research will use ethnographic methods to explore how conversations about blood tests influence testing practices in primary care. By observing real-world interactions between clinicians and patients *in situ,* this project aims to identify ways to improve blood test communication and support more prudent, patient-centred testing practices.

2. Why is this research important?

This research is the first to use ethnographic methods to explore how blood tests are discussed in GP settings. The project will open the 'black box' of blood test discussions by directly observing communication between GP staff and patients, providing important insight into how communication can sustain or challenge inappropriate blood test use in the GP setting. Through the use of innovative, participatory methods, GP staff and patients will be invited to identify practical solutions to improve blood test communication.

This research is timely. The NHS Long Term Plan stipulates that all patients must have access to their online records – including test results – by the end of 2024,[8] a development which has significant implications for blood test communication. Previous research shows that that online access can result in unintended consequences, including patient anxiety about blood test results and increased workload.[9,10] This project will explore these issues in-depth, providing critical insight into how digital access has shaped blood test communication.

Ensuring that blood tests are used wisely is important for the sustainability of primary care. Primary care clinicians request 45% of all NHS blood tests, at a cost of approximately £1.8bn in 2015-16.[1] Unnecessary blood tests can lead to costly follow-up care such as outpatient visits and hospital investigations.[11] By encouraging GP staff to speak more clearly about the limitations of blood tests, this research can help to reduce overuse and lead to much-needed cost-savings for the NHS.

The project will also seek to alleviate the pressures faced by GP staff. A survey by the Royal College of General Practitioners (RCGP) found that three-quarters of GPs feel patient safety is being compromised by excessive workloads,[12] and the British Medical Association has warned of 'a growing crisis' in primary care.[13] By identifying strategies to improve blood testing practices, this research will

free up GP consultation time and reduce the administrative burden associated with managing blood test results.

This work aligns with NIHR's 'Prevention and Public Health' priority research theme for 2024-25 by seeking to understand how blood test investigations can be used in primary care to provide maximum benefit with the least harm. The project will build on a number of NIHR-funded projects on primary care blood tests by providing rich ethnographic insights into how communication shapes blood testing practices.

3. Review of existing evidence

The volume of blood testing in primary care is a financial and logistical burden, and many of the blood tests ordered in primary care appear to be of little diagnostic or prognostic value. In the Why Test study, which reviewed blood test results for 2572 patients across 57 UK GP practices, only 6.2% of tests led to a new or confirmed diagnosis, while nearly half of tests (48.8%) did not result in patient reassurance or any change in management. Furthermore, a quarter of tests were felt to be partially or fully unnecessary when retrospectively reviewed by another clinician.[3]

Increased blood testing exposes patients to the risk of overdiagnosis, which causes harm through further investigations, treatments, or anxiety. The harms of blood tests may be overlooked in comparison to more invasive diagnostic procedures, but blood tests are not benign interventions. Borderline results are common – in one study of GP blood tests, only 26.6% of patients had completely normal blood test results[3] – and can lead to unnecessary anxiety, particularly if patients access these results via online apps.[10] False-positive or borderline results can also lead to 'investigation momentum': further testing or invasive procedures such as biopsies and irradiating scans.[14] Alternatively, normal blood test results in a patient presenting with symptoms may provide undue reassurance which delays diagnosis, or leads focus away from discussing lifestyle changes.[15]

Despite the limitations of testing, patients tend to have high expectations for blood tests as a diagnostic tool[18,19] and may underestimate the potential harms.[20] Patients may request blood tests as part of a general 'check-up'[18] despite limited evidence for the value of testing in healthy asymptomatic individuals.[21,22] A UK-based qualitative study by Watson *et al.* interviewed GPs and patients at the time of blood testing and again after test results were received, finding that there was often a mismatch in patients' and GP's expectations of testing.[4] Patients felt that blood tests would provide answers to and solutions for their symptoms, and perceived the offer of testing as a sign that the doctor was taking their concerns seriously. By contrast, GPs tended to be less optimistic about blood tests, viewing them as a way to rule out serious problems rather than pin down specific diagnoses. Even if they feel blood tests are unlikely to be of clinical use, clinicians may acquiesce to patient requests in order to avoid lengthy discussions about the merits of testing.[15]

Communication between clinicians and patients appears to be an important driver of medical overuse. Research into antibiotic prescribing practices in primary care, for example, suggests that clinicians often mistake expressions of concern about symptoms as requests for antibiotic treatment.[16] Similar patterns may underlie overuse of blood tests, as in research which suggests that clinicians may initiate testing to satisfy or affirm patients even where there has been no explicit patient request.[17]

Although there is a wide body of research using directly-observed data to examine communication in primary care consultations,[25–30] there is only one UK-based study of this kind which focuses on blood test discussions: a secondary content analysis of 27 audio-recordings from an archive of GP consultations, which showed that shared decision-making was rarely evident in blood test discussions.[6] While the study quantified various aspects of poor communication – for example, that patients were given the opportunity to ask questions about blood test results only 13% of the time – there remains a gap in the literature for a more detailed and in-depth account of blood test communication using naturalistic data. This will help us to better understand the interactional factors that contribute to increased testing rates, as well as the difficulties patients report in understanding their blood test results.

4. Aims/research questions

Aim

To understand how GPs and other primary care clinicians discuss blood tests with patients, and how this communication might be improved to support appropriate use of blood testing in primary care.

Research questions

1. How is the issue of blood testing communication framed in the existing literature?
2. What do clinicians and patients seek to accomplish with blood tests and how is this communicated?
    a. How are decisions about ordering blood tests negotiated and communicated in practice?
    b. How do clinicians and patients communicate what they understand about blood test results?
3. What concerns do primary care clinicians have about blood testing communication, and how can insights from participatory ethnography support clinicians to identify solutions to these concerns?
4. How can participatory creative methods be harnessed to support more effective communication between patients and primary care clinicians about blood tests?

5. Project plan

Phase 1: understanding blood test communication in primary care

Work Package (WP) 1: Synthesising existing qualitative literature about primary care blood testing discussions.

Objectives

1. Identify when, where and how blood tests are discussed in primary care.
2. Explore how patient and clinicians' perspectives on and experiences of blood testing are framed in the existing literature.
3. Develop new conceptual understandings of blood test communication in primary care settings.

Methods

I will conduct a meta-ethnography of qualitative literature exploring communication about blood tests in a primary care setting. Meta-ethnography is a form of interpretative qualitative synthesis, involving detailed re-interpretation of the primary data and themes in published literature.[31] Unlike other aggregative qualitative synthesis approaches, meta-ethnography aims to develop new theories about the topic under study.[32] Meta-ethnography is particularly useful for synthesising small numbers of papers, and I have identified <20 qualitative studies on this topic.

I will follow the seven stages of meta-ethnography described by Noblit and Hare: (1) getting started; (2) deciding what is relevant; (3) reading the studies; (4) determining how studies are related; (5) translating the findings; (6) synthesising translations; (7) expressing the synthesis.[33]

Search strategy

I will conduct a systematic literature search of medical, social science and public health databases, using appropriate search terms and operator functions as guided by a QMUL research librarian. I will also search grey literature via thesis repositories, Google Scholar, citation searches, and reference list reviews. In order to keep the review relevant to contemporary practice, I will restrict papers to those published (in English) within the past 15 years.

Studies will be imported into Mendeley Reference Manager. After removing duplicates I will screen the titles and abstracts of the remaining papers against SPIDER criteria:[34]

- *Sample:* staff members working in primary care; patients accessing these services.
- *Phenomenon of Interest:* blood testing
- *Design:* interview, focus groups, case studies, ethnography.
- *Evaluation:* motivations, thoughts, beliefs, understandings.
- *Research type:* qualitative (including mixed-methods).

I will screen the full texts of selected papers from the first round of screening. A random sample of 50% of these papers will be double-screened by a second reviewer (e.g. Professor Swinglehurst). We will meet to discuss and resolve any disagreements.

Data extraction

Study characteristics will be extracted and imported into a shared spreadsheet. The second reviewer and I will undertake quality appraisal of each study using Joanna Briggs Institute (JBI) criteria for qualitative research.[35]

Through repeated readings, the second reviewer and I will extract data from each study, capturing themes, first-order constructs (participant quotes) and second-order constructs (the study authors' interpretations).

Data analysis and synthesis

I will work with a second reviewer throughout the complex analytical stages (4-6) of the meta-ethnography. We will hold analysis sessions at each stage with qualitative researchers within the QMUL APOLLO team to ensure rigour.

In stage 4, relationships between studies will be identified by iteratively grouping related concepts and themes. In stage 5, findings will be translated by mapping concepts from one study to another, using a constant comparative method to identify where findings converge or diverge. Finally, in stage 6, shared themes and concepts from across studies will be integrated into third-order constructs to develop a 'line of argument'.[31]

When writing up the findings of the meta-ethnography I will use the eMERGe reporting framework developed by France *et al.*[36]

WP2: Understanding blood testing practices and interactions in the primary care setting.

To use ethnographic methods to explore in-depth:

1. How decisions to do blood tests are negotiated between patients and GP clinical staff.
2. The role of language and non-verbal communication in interactions about blood tests.
3. GPs' decision-making processes regarding blood test results.
4. Patients' experiences of using digital tools to access blood test results.
5. How and whether uncertainty around blood tests is communicated.
6. The approaches used by GP staff to convey and discuss blood test results.

Methods

I will conduct an ethnographic study of blood testing practices and communication in GP practices. Ethnography is a qualitative method involving immersion in a particular setting through observation, interviews, reflexive diaries and fieldnotes. It is increasingly recognised as a useful method for primary care research.[2,37–40]

Ethnography provides valuable insights into the everyday practices of clinicians and patients, revealing practices, beliefs and behaviours which may be taken for granted or viewed as unremarkable.[41,42] It can uncover the complexity in routine interactions such as blood test discussions by capturing meaningful details that people might not remember or mention in interview studies.[43] Ethnographic approaches have been used in previous research to develop new understandings of how medical overuse arises and is sustained through social interactions, norms and practices.[41,44,45]

Sampling

Observation will take place at two different GP practices to facilitate comparison whilst still allowing for a long period of in-depth observation in each site. Participants in PPI workshops felt it was important to explore blood testing practices in non-urban settings and in diverse patient groups. I will therefore recruit an inner-city practice with a young, multi-ethnic population and a suburban practice with an older patient population. I will use General Practice Fingertips data to identify potential research sites.[46]

Data collection

I will undertake observation in each site 2-3 times per week over 4-6 months, conducting approximately 200 hours of observation. Modes of data collection will include:

1. *Observation of professional practices*
    - This will include clinical consultations, reception/administrative areas, interprofessional meetings, and Patient Participant Groups. To capture the full range of testing practices I will observe consultations with phlebotomists and nurses as well as GPs. I will make detailed fieldnotes during and after observations.

2. *Opportunistic ethnographic interviews with patients and GP staff*
    - These informal interviews may be recorded (with consent) using a video recorder, or will form part of my fieldnotes.
    - In order to capture digital aspects of communication about blood tests, I will conduct ad-hoc think-aloud interviews with patients and staff, asking interviewees to describe their thoughts and impressions as they use digital resources.[2,47]

3. *Gathering documents from electronic records which relate to blood testing* (e.g. consultation notes detailing discussions about blood tests, screen captures of patient apps).

4. *Recording consultations*
    - I will record approximately 45 consultations/think-aloud interviews in which blood tests are discussed. To ensure diversity of the dataset I will ensure to include consultations where interpreters are used. I will aim for a spread of recordings across the testing process:
        i. discussing/ordering tests;
        ii. receiving/filing blood test results;
        iii. communicating results to patients.

Consent

I will seek consent from clinicians and patients to record consultations either prospectively or retrospectively (e.g. if seeking to access audio consultation data from practice telephone systems). Participants in PPI workshops were surprised that many GP practices automatically recorded consultations and felt patients needed to be told this before being approached for consent. I will consider this in the consent procedure, guided by the PPI panel and in collaboration with research sites.

Phase 2: Improving blood test communication in primary care

WP3: Exploring and addressing professional concerns around blood test communication.

1. Elicit the perspectives of primary care staff on communication about blood tests.
2. Support primary care staff to develop new ways of managing blood test communication.

Methods

Video-reflexive ethnography (VRE) is an innovative participatory research method. It involves filming people *in situ* doing their usual work, and then showing short clips to relevant actors in facilitated

workshops.[48] Participants in VRE workshops watch these recordings to see familiar practices from a fresh perspective, stimulating discussions about what they may have taken for granted and where they might see possibilities for change. VRE therefore directly engages those delivering care in the process of data analysis and quality improvement.

VRE has been used to improve practice in various complex healthcare settings, including maternity services, [49], acute medical units,[50] and dementia care.[51] My supervisors Professor Swinglehurst and Dr Fudge have successfully used this methodology to explore structured medication reviews in GP practices.[52]

VRE workshops will be conducted according to the four key principles laid out by Iedema *et al.:* [48]

1. *Exnovation:* acknowledging the complexity of existing practices and encouraging innovation with existing resources.
2. *Collaboration:* valuing the experiences and expertise of participants and supporting a process of co-creation.
3. *Reflexivity:* encouraging participants to re-evaluate their assumptions, beliefs and practices.
4. *Care:* ensuring that participants and researchers feel safe and supported.

Data collection

I will hold six one-hour workshops (three per site), each involving 6-10 primary care staff members. I will work with practice managers/GP partners to schedule these workshops in time set aside for training in order to minimise disruption to clinical activities.

In each workshop I will show selected short recordings from WP2 to prompt critical discussions about blood testing conversations in primary care. These discussions will be audio-recorded and transcribed for analysis. VRE workshops and analysis will take place concurrently, allowing findings from one workshop to inform the selection of recordings for the next.

The focus of the workshops will be on blood test communication as a 'critical moment' for ensuring appropriate use of primary care resources and facilitating patient involvement in their own healthcare. They will provide an opportunity for primary care professionals to explore alternative approaches to blood test communication in a safe and supportive environment.

WP4: Co-designing strategies to improve blood test communication.

1. Co-create an interpretation of study findings with the PPI panel and theatre-makers.
2. Bring together members of the public and primary care staff to collectively develop strategies to improve blood test communication.
3. Co-produce a workshop resource (for primary care staff) and public engagement materials to help improve blood test communication and develop patients' understanding of blood tests.

Methods

Forum Theatre is a participatory performance methodology which allows participants to directly engage in enacting, critiquing and transforming real-world scenarios. It is one of the tools for social change described in Augusto Boal's *Theatre of the Oppressed*, developed in the 1960s and 1970s whilst working with marginalised communities in Brazil.[53]

In a Forum Theatre workshop, actors stage a scene involving conflict or misunderstanding while audience members (or "spect-actors") observe and reflect on the social dynamics which emerge.[54] A facilitator then invites spect-actors to suggest alternative actions or responses to change the scene. An improvised version of the scene then takes place, with spect-actors able to intervene at any time by asking actors to change their behaviour or stepping into the scene themselves.[54]

Forum Theatre is increasingly used in healthcare research as a way to explore and change interpersonal dynamics within healthcare settings.[55–57] Like VRE, Forum Theatre provides a space for participants to see situations from alternate perspectives, inviting them to consider how practices and ways of working might be transformed.

Co-production process

Part 1
A day-long workshop with the PPI panel and actors from Improbable, an improvisation theatre company. Using selected research materials (e.g. analytic themes from the meta-ethnography, ethnographic observations, VRE workshop excerpts), workshop participants will co-design a short play with multiple scenes, based on the study findings.

Part 2
Two Forum Theatre workshops, each involving approximately 20 local primary care staff and members of the public, who will be recruited in collaboration with Social Action for Health (a local community health charity – see Collaborations). The workshops will be led and facilitated by actors from Improbable, who will perform each scene from the short play in turn.

The play will be professionally recorded, including both the 'original' scenes and the adaptations created in the forum theatre workshops. Scenes in which audience members have participated will not be used in further outputs except with a participant's written consent.

Part 3
In two further workshops I will work with the PPI panel to adapt the play into two resources to disseminate study findings and support improved blood test communication: a workshop for primary care staff and an online publicly-accessible resource for patients. Both resources will incorporate clips from the forum theatre exercise, using these as prompts for reflection. I will also work with the PPI panel to develop and review other public dissemination materials like posters and social media posts.

Analysis

This study will produce a rich and varied dataset including fieldnotes, audio- and video-recorded consultations, VRE workshop transcripts, and reflexive diaries. In my analysis I will draw upon linguistic ethnographic approaches, combining insights from ethnographic observation with close analysis of how language is used *in situ* by patients and GP staff.[58] Linguistic ethnography offers unique insights into the interpersonal dynamics involved in working practices. Widely used in education research, it is increasingly recognised as a valuable tool for understanding the role of language in medical interactions, including in primary care. [37,45,59]

NVivo 15 will be used to manage study datasets and CLAN software will be used to support the transcription of recorded consultations and VRE workshops. To allow me to conduct linguistic microanalysis of consultation data I will use Jeffersonian notation (following Conversation Analysis conventions) to capture fine-grained interactional details such as pauses and overlaps.[60] I will use an inductive approach to identify 'critical moments' from consultations to be shared in VRE workshops.

6. Dissemination, outputs and anticipated impact

Dissemination of key outputs

This project will produce two main outputs, both adapted (with help from the PPI panel) from the short play co-produced in WP4:

Workshop resource for clinicians:

- I will deliver the workshop in teaching sessions for local GP practice staff and GP VTS schemes, and will seek early feedback to refine the workshops and demonstrate impact.
- To ensure the resource reaches clinicians nationally, I will run the workshop at GP conferences.
- I will share the resource through networks for academic GPs and with the Programme Directors of GP VTS training schemes across the country.

Online resource for patients:

- This will be hosted on the project webpage and YouTube.

- It will be promoted to the public via social media (e.g. X), flyers at local community groups and GP practices, and patient participant group mailing lists.
- Key messages from the Forum Theatre workshops will be disseminated to patients in written form via leaflets and newsletters.

Dissemination of study findings

To patients/the public:

- I will maintain a project webpage on the APOLLO team website throughout the fellowship and beyond, on which PPI panel members will share summaries of the research plan and findings.
- I will write blogs to be shared on the project website and provide project news via an X (formerly Twitter) page.
- Guided by the training I will receive in public engagement methods and science communication, I will produce a podcast for APOLLO team's podcast, *Thinking In Between.*
- I will host a stall at QMUL's Festival of Communities, an annual public engagement event to showcase research from QMUL and local organisations (with >25,000 attendees since its inception in 2016).

To all stakeholders:

Together with the PPI panel, and with the support of the QMUL Centre for Public Engagement, I will host a public dissemination event for local clinicians, community groups and the public. The aim of this event is to share the research findings and discuss how the resources developed in the project might be shared to improve blood test communication in primary care.

To academics/clinicians:

- Publication of at least four peer-reviewed open-access papers in target academic journals:
    - Meta- ethnography *(Sociology of Health and Illness)*
    - Linguistic ethnography *(Social Science and Medicine)*
    - Video-reflexive ethnography *(BJGP)*
    - Co-producing resources to improve blood testing communication *(BMJ Evidence-Based Medicine)*
- Presentations at national conferences (e.g. Society for Academic Primary Care and the Royal College of General Practitioners).
- Presentations at Preventing Overdiagnosis (an multidisciplinary international conference well-attended by primary care clinicians) and WONCA (World Organization of Family Doctors).
- Presentations in clinical and academic networks e.g. RCGP Optimal Testing, University of Bristol Centre for Academic Primary Care, and Wolfson Institute of Population Health.

Anticipated impact

This research will:

i. Prompt lay and professional discussion about the challenges of blood testing and how they can be overcome;
ii. Reframe the primary care consultation as a key site of opportunity for the safe and prudent use of blood testing;
iii. Develop a 'good practice' approach to discussing blood tests in primary care, co-created with lay participants.

Key impacts include:

For patients:

Patients will feel more confident making shared decisions about whether or not to do blood tests, will have a greater insight into how primary care clinicians approach blood testing, and will understand their blood test results better.

For primary care staff:

GPs and allied health professionals will feel more confident in supporting patients to make decisions about doing blood tests and in explaining their results.

For the NHS:

Improving blood test communication and shared decision-making between clinicians and patients is likely reduce overall numbers of blood tests. This will reduce primary care blood testing costs and lessen the administrative burden on primary care staff. It will also reduce the costs associated with repeat and follow-up investigations.

7. Project management

I will be responsible for the project's progress, budget and outputs, with support from my supervisors and the QMUL finance team. I have prepared a research timetable (see uploads).

The progress of the project will be monitored through:

Supervision

- Face-to-face meetings with Professor Swinglehurst every fortnight and with Dr Fudge once a month.
- Monthly online meetings with Dr Watson, with twice-yearly in-person visits in Bristol.
- Whole supervisory team meetings three times a year, with at least one face-to-face meeting in London per year.
- All supervisors will provide support over email as required and offer feedback on written work.

Project Advisory Group

I have assembled a Project Advisory Group (PAG) which will meet online bi-annually to assess and guide study progress. PAG members will also act as mentors on an ad-hoc basis. The PAG will be co-chaired by two lay members, Steven Oliver and **name**, who helped to develop this proposal. Alongside my three supervisors, there will be four other PAG members with relevant expertise:

- Professor Natalie Armstrong (Professor of Healthcare Improvement Research, University of Leicester)
    - Medical sociologist with expertise in applied qualitative research and overdiagnosis.
- Dr Julian Treadwell (GP & NIHR Academic Clinical Lecturer, University of Bristol)
    - Lead for GP Evidence, a co-produced information resource about the benefits and harms of treatments for long-term conditions.
- Dr Margaret McCartney (GP, writer and Senior Clinical Lecturer in General Practice, University of St Andrews)
    - Internationally-recognised advocate for evidence-based medicine, with extensive broadcasting experience.
- Dr Jessica Drinkwater (GP & NIHR Academic Clinical Lecturer, University of Manchester)
    - Academic GP with wide-ranging experience of using participatory and applied theatre methods to strengthen patient involvement in primary care service improvement.

Formal reviews

In line with QMUL university regulations, I will have regular reviews to assess my progress and identify any support or training needs:

- Progression review at 6-9 months
- Annual progress reviews at 12, 24, and 36 months

8. Ethics

I will obtain ethical approval from QMUL and the NHS Health Research Authority prior to carrying out any data collection involving participants. I will take particular ethical precautions with the following aspects of the project:

Ethnographic observation and recording
I will seek written consent from each member of staff within the GP practice to observe them in their usual role. I will conduct all observations respectfully and as unobtrusively as possible, leaving a space immediately if I am asked to do so by staff. In line with usual practice for students observing patient care, GP practice staff will seek a patient's explicit consent for me to observe their care, ideally without me present (e.g. as they collect the patient from the waiting room).[61]

Video-reflexive ethnography workshops
Staff may feel uncomfortable watching their consultations or be upset by comments made about their practice by other participants. This risk will be mitigated by ensuring that GP staff are aware of the purpose of the workshops: i.e., to collectively identify new ways of working, rather than critiquing each other's practice. Professor Swinglehurst and Dr Fudge will support me in planning the workshops and developing a plan for facilitation.

Confidentiality
This project involves the collection of video- and audio-recorded data of clinical consultations as well as workshops involving GP practice staff. Ensuring the confidentiality and security of data is thus essential; this was also a high priority for PPI participants. All research data will be securely stored in line with QMUL's data management policy, the Data Protection Act 2018, and the conditions of the Research Ethics Committee approval. Recorded data will be securely stored in encrypted form and then deleted from the recording device.

9. Success criteria

I will measure the success of my PhD according to the completion of the milestones described in the attached Gantt chart. The following indicators will be particular measures of success:

1. Ethical approval in place by end of year 1;
2. Meta-ethnographic synthesis submitted for publication by 1st quarter of year 2;
3. Ethnographic data collection and VRE workshops complete by end of year 2;
4. Co-produced resources for patients and clinicians finalised by 3rd quarter of year 3;
5. Application for postdoctoral funding (to scale up and evaluate the resources co-produced in WP4) in-progress by the end of year 3.

Risks and mitigation

1. Delays in obtaining ethical approval: I will apply for ethical approval promptly upon starting the fellowship. I will be supported by my supervisors in preparing a high-quality application to reduce the risk of the initial application being refused.

2. Difficulties completing the meta-ethnography: I have done a rapid scoping search and ascertained that there is sufficient relevant qualitative literature for a meta-ethnographic review. I will work with a QMUL librarian to prepare the search strategy and have identified two relevant training courses. Other members of the APOLLO team will be able to offer methodological support.

3. Difficulty gaining consent to record consultations: The majority of patients in previous research gave their consent to the video-recording of GP consultations for research purposes.[62] Participants in pre-proposal PPI workshops also felt that recording consultations would be acceptable to patients. I will use the period of ethnographic observation to build trust with GP staff prior to making any recordings, and will seek the PPI panel's guidance on recruitment materials including consent forms.

4. Challenges in the applied theatre workshops: I anticipate that the forum theatre approach in WP4 will be new to most of the participants and that some may feel unsure about the aims or process. This will be mitigated by the expertise of the Improbable actors, who are experienced in developing improvised theatre productions in collaboration with audiences. I will be supported by my mentor, Dr Drinkwater (on the Project Advisory Group), who has experience of using applied theatre methods.

PPI in proposal development (349/350 words)

To decide my research questions and develop this proposal, I held three workshops with 18 members of the public who had recently had general practice blood tests. Participants were from diverse ethnic backgrounds, including Black British, South Asian and Chinese, and ranged in age from 20s-70s. Several had long-term conditions and/or identified as disabled, and two participants were neurodivergent.

In the first workshop I asked about participants' experiences of blood testing, which helped shape my research questions. In later workshops, we discussed the proposed research: the topic, aims and objectives, methods, and plans for sharing findings. Participants felt strongly that this research was important. They highlighted communication difficulties throughout the blood testing process:

- *"I don't feel the GP discusses the decision to do a blood test… they just say you need one."*
- *"There's a lot of difficulty in understanding the results without having to go onto Google."*
- *"The doctor says 'we don't worry when [the blood test] is just out of the range', but there's no time for the GP to properly explain what that means."*

Participants thought that spending several months observing in each general practice would help me develop trust with staff and patients, making everyone more comfortable with recording consultations. They liked the idea of using applied theatre methods and felt that a play would be an accessible way of sharing the research.

Based on feedback from these workshops, I made the following changes to the project:

- I will observe Patient Participation Groups (PPGs) as part of the ethnography and share my findings in PPG meetings to involve existing patient networks in the research.
- Participants wanted the workshop resource for GPs to be part of GP training, so I will run the workshop for local GP training programmes.
- I decided not to include GPs in the workshop session to develop the play, as participants felt that PPI contributors would be able to participate more freely without staff present.

Two workshop participants co-wrote the plain English summary and provided feedback on the PPI plans. They will co-chair the Project Advisory Group.

PPI in proposed research (348/350 words)

I have assembled a Patient and Public Involvement (PPI) panel who will be involved throughout the project. The panel is made up of people who have had a blood test in primary care in the past year. People on the panel are diverse in age, ethnicity and gender and have a range of health experiences.

I will meet with the panel three times a year to discuss study progress and troubleshoot any issues which come up. I will seek their guidance on recruiting research sites and obtaining consent for recordings. I will also gather their feedback on early findings from general practice observations. This will help me choose recordings to show in video-reflexive ethnography workshops.

The PPI panel will work with a theatre company to develop a play based on the research findings, which will used in theatre workshops with members of the public to find ways to improve blood test communication. I will recruit participants for the workshops in collaboration with Social Action for Health, a local charity with extensive experience of working with East London communities who are not well-represented in research.

I will collaborate with the panel to adapt the play into a workshop (for clinical staff) and a public-facing online resource (for patients). These resources will support primary care staff and patients to communicate better about blood tests. The panel will also develop summaries of the research plan and findings to post on a project webpage. At the end of the project, the PPI panel will help to host a public event to share our findings with local clinicians and members of the community.

All panel members will be paid for their time and travel expenses in line with NIHR guidance, including pre-meeting work. I will also offer payments for childcare/caring expenses. The research budget

includes funding for panel members to access research training and to present the research at conferences.

Two lay participants helped to develop this proposal. They will co-chair a project advisory group with my supervisors and four external academics/GPs. This group will provide scrutiny, guidance and oversight throughout the project.

Training and development plan (1006/1000 words)

The QMUL Doctoral College and the Wolfson Institute of Population Health offer a doctoral development programme with courses covering project management, personal effectiveness, and research governance. I will attend induction events, yearly cohort days, and research skills seminars and workshops. As outlined below and in the attached training timetable, I have planned further training to allow me to successfully complete the PhD and continue my development as a clinical researcher.

Research skills

Qualitative methodology:

- Advanced Qualitative Research Methods (8 weeks part-time, University of Oxford, £3175): Masters module focusing on more complex aspects of qualitative research such as reflexivity, bias, and advanced data collection methods. This will underpin the qualitative methods training I received in my MSc and MPhil.

Meta-ethnography (WP1):
To refresh my skills in systematic review and develop my understanding of qualitative synthesis, I will take the following courses:

- Literature Searching for Systematic Reviews (half-day, QMUL): course on literature searching in preparation for the meta-ethnography database searches.
- Overview of Qualitative Review and Synthesis Methods (1 day, University of Bristol): course to aid me in conducting an interpretive qualitative synthesis.

Ethnography and video-reflexive ethnography (WPs2&3):
Building on my previous experience of ethnographic research, I will take the following external courses:

- Ethnography and Ethnographic Methods (5 days, Essex Summer School in Social Science Data Analysis, £1,600): module to provide a thorough introduction to the theory and practice of ethnographic research and set me up for ethnographic observation in GP practices.
- Summer school module on Linguistic Ethnography (5 days, Ghent University, £250): an introduction to key concepts in linguistic ethnography. This will support me in the close analysis of recorded data in WP2, building a Conversation Analysis course (University of Oxford) I completed in 2024.

Creative co-production (WP4):
I will be mentored by Dr Jessica Drinkwater, a member of the PAG who has experience of using forum theatre to improve primary care services. I will also take the following courses:

- Creative Methods in Qualitative Research (2 days, Social Research Association, £220): course exploring creative methods for data collection and analysis.
- Forum Theatre Intensive Training (5 days, School of Theatre of the Oppressed, £850): intensive course to develop skills in applied theatre methods.

Dissemination and impact

Patient and public involvement/engagement:
To support me in working with public contributors throughout the project I will take the following courses:

- Co-design, Co-production and Participatory Action Research; Inclusivity within Research Processes (1 day each, University of Bristol)
- Hands on Engagement: Taking Research to Events and Festivals (1 day, QMUL): course to prepare me to host the dissemination event.

Implementation of findings:

- Knowledge into Action (8 weeks part-time, University of Oxford, £3175), led by Professor Carl Heneghan. This module will support my plans for impact and dissemination, helping me to understand how my recommendations can be implemented in clinical practice. It will also inform an application for postdoctoral research to develop and test interventions to improve blood test communication.

Disseminating findings:

- Science Communication Masterclass (4 days, UWE Bristol, £400) : intensive course on communicating research findings with the public. This will help me in sharing my PhD research via podcasts, blogs and social media.
- WordPress website design (1 day, CityLit, £129) – course to help me design a project website to share study findings and patient resources.

Conferences:
I will attend and present work at conferences including:

- Royal College of General Practitioners
- Society of Academic Primary Care
- World Organization of Family Doctors (WONCA)
- Preventing Overdiagnosis: an international conference with a strong primary care presence

Development as a researcher

Leadership:

- Emerging Clinical Leaders (6 days in total, King's Fund, £2712): interprofessional programme to build skills in leadership, resilience and team management. This will further my development as an independent research leader.

Teaching:

- To develop as an educator I will facilitate undergraduate/Masters module seminars within the department when my research commitments allow. I will attend Graduate Teaching Assistant Development workshops provided by QMUL.

Research networks:
I will be a member of the following research networks, allowing me to learn from other researchers, get feedback on my work, and develop new collaborations:

At QMUL:

- APOLLO Social Sciences team:
    - Multidisciplinary team of qualitative primary care researchers, led by Professor Swinglehurst. Alongside regular team meetings I will have the opportunity to attend APOLLO's termly writing retreat.
- Thinking Between the Lines:
    - Qualitative research network, co-convened by Dr Fudge, holding monthly meetings for researchers to present work-in-progress and receive feedback.

Externally:

- Preventing Overdiagnosis Early/Mid-Career Researcher Network:
    - International network of researchers working on overdiagnosis, which provided helpful feedback on this proposal.
- RCGP Optimal Testing group:
    - Subgroup of the RCGP Overdiagnosis interest group, led by Dr Watson. I will attend their tri-yearly meetings to share my work and discuss how to improve primary care blood testing practices.

Research visit

In the 3rd year of the PhD, I will spend six weeks in Australia, visiting two institutions which are pioneering work on communicating overdiagnosis to patients and developing patient education resources. In each site I will be mentored by the unit lead and a mid-career researcher. We will agree a timetable in advance of my visit.

- Sydney Health Literacy Lab, University of Sydney
    - The Lab focuses on healthcare communication to prevent overdiagnosis and low-value care. I will spend four weeks here, hosted by Professor Carissa Bonner, a behavioural scientist who specialises in health literacy and risk communication, and Dr Brooke Nickel.
    - I will use insights from the Lab's extensive work co-producing health literacy tools to aid the development of the WP4 resource for patients.
- Australian Centre for Health Engagement, Evidence and Values (ACHEEV), University of Wollongong.
    - I will spend two weeks at ACHEEV, hosted by Professor Stacy Carter and Dr Patti Shih. The ACHEEV team aims to find solutions to medical overuse by making evidence more accessible.
    - I am a technical advisor for Dr Shih's project developing an online resource to explain blood test results to Australian patients, and will be able to develop this collaboration during my visit.

Supervisor (250 words per supervisor)

Professor Deborah Swinglehurst (250/250)
Deborah Swinglehurst is a Professor of Primary Care and general practitioner. Her academic work uses qualitative methods, including ethnography and linguistic ethnography, to explore how interaction and social relationships shape healthcare practices. She leads the APOLLO social science group, adopting innovative methodologies at the interfaces of medicine, social science and linguistics to address complex healthcare challenges.

She is Institute co-lead for Applied Social Science within the QMUL Primary Care Unit and sits on the Executive Board and Research Committee. She is in the senior management team for the Wellcome-funded Doctoral Training Programme on Health Data in Practice and the leadership team for QMUL's NIHR SPCR membership.

Professor Swinglehurst has published widely in journals including *BMJ, BMJ Quality & Safety, BJGP*, and *Social Science and Medicine*. She is on the 2024 Stanford/Elsevier Top 2% Scientists list of the most cited scientists globally, and was awarded the 2019 RCGP/SAPC John Fry Award for her contribution to general practice. Her recent NIHR-funded research on multimorbidity and polypharmacy involved co-producing a suite of RCGP e-learning modules (reaching >1000 learners) and a public engagement resource (*Let's Talk Differently about Medicines*) which has reached >1800 people.

She will provide expertise in meta-ethnography, ethnographic approaches (including linguistic ethnography/video-reflexive ethnography), and creative co-production. We will have supervision meetings fortnightly and she will provide further support as needed.

Accredited by the UK Council for Graduate Education as a recognised research supervisor, she has supervised 6 PhD students to completion, currently supervises 6 PhD students, and has examined 10 PhDs.

Dr Jessica Watson (250/250)
Jessica Watson is a general practitioner and NIHR Academic Clinical Lecturer whose research uses mixed methods to understand and improve primary care blood testing practices. She chairs the Primary Care Academic Collaborative and leads the Optimal Testing division of the RCGP Overdiagnosis group.

Dr Watson has been awarded three consecutive NIHR Fellowships and has delivered over £2m of research funding. She has authored 64 peer-reviewed research articles which have been cited over 2,100 times. Her work has been featured on BBC radio, ITV news, and in *BMJ, BJGP* and *RCGP* podcasts. She was Principal Investigator on the NIHR-funded Test Talk Tools study (2022-2024), and currently leads two further NIHR-funded studies:

- Accessible Results (Principal Investigator): qualitative project developing tools to help patients with diverse needs access and understand online blood test results.
- Optimal testing for long-term condition monitoring in primary care (Joint Lead Applicant): five-year mixed-methods study to develop, implement and evaluate a new blood test monitoring strategy for long-term conditions.

Dr Watson will provide expertise in blood testing and overdiagnosis as well as mentorship in research dissemination. She will offer opportunities to collaborate with researchers at the University of Bristol and within PACT. We will meet online monthly and I will travel to Bristol for in-person supervision twice-yearly. We will have joint online meetings with Professor Swinglehurst and Dr Fudge three times a year to ensure regular contact between all supervisors.

She is currently supervising 1 PhD student. She has supervised 2 GP ACFs and 3 MSc students.

Dr Nina Fudge (250/250)
Dr Fudge is a Lecturer in Social Science and THIS Institute Research Fellow with a background in social anthropology. Her research explores knowledge, expertise and practice in complex healthcare settings such as community pharmacies and GP practices. Her recent research focuses on patients'

and clinicians' experiences and practices of polypharmacy, aiming to develop safe and person-centred approaches to primary care medicines use.

She has extensive experience of qualitative participant-focused methods, including narrative, ethnographic and visual approaches. She has particular expertise in video-reflexive ethnography (VRE) which she has used in multiple polypharmacy research projects, including her THIS Fellowship (awarded 2021) which uses VRE to improve the delivery of structured medication reviews.

She has worked closely with Professor Swinglehurst since 2017 as part of the APOLLO-MM project, including on the co-production of patient-facing resources to encourage discussions around polypharmacy. She is Deputy Director of Graduate Studies for the Wolfson Institute of Population Health and co-convenes Thinking Between the Lines, a qualitative research network through which I will receive feedback on my work and take part in data analysis workshops.

I will meet with Dr Fudge on a monthly basis throughout the fellowship. She will provide particular guidance on hosting the VRE workshops in WP3 and analysing the resulting data. With her experience of co-design methods she will also support me in co-producing the clinician and patient resources in WP4.

She is currently supervising 3 PhD students (1 with Professor Swinglehurst) and has supervised 3 PhD students to completion (2 with Professor Swinglehurst).

Collaborations (627/600 words)

Academic collaborations

Centre for Academic Primary Care (CAPC), University of Bristol

Through Dr Watson I will develop links with CAPC and their ongoing blood testing research, including the Accessible Results study – a participatory study which explores how people with diverse needs can be supported to access blood test results online. Through working with CAPC I hope to foster cross-institutional collaboration on a project Professor Swinglehurst is developing which explores GP blood testing organisational routines.

Primary care Academic Collaborative (PACT)

PACT is a UK network of primary care researchers which has run several studies relevant to the proposed PhD: the Why Test study, investigating the indication for primary care blood tests; and the Hidden Workload study, exploring primary care clinician workload. I hope to collaborate with PACT to disseminate the research findings, including the workshop for GP staff, and will explore the possibility of developing a spin-off project exploring blood testing communication nationally.

Sydney Health Literacy Lab/ Australian Centre for Health Engagement, Evidence and Values

I will collaborate with these two centres which work in partnership as part of Wiser Healthcare (https://www.wiserhealthcare.org.au/), a research initiative to reduce overdiagnosis in Australia and worldwide. The research visit will allow for a mutual exchange of methodological approaches. The Sydney Health Literacy Lab hosts a number of behavioural scientists from whom I hope to learn about implementation science and resource development. At ACHEEV, I will learn about their empirical ethics work using community juries to define and make judgements about overdiagnosis. In return, I will be able to share my experience of using ethnographic methods to understand how overdiagnosis emerges in practice.

Clinical collaborations

Dr Stefán Hjörleifsson, Choosing Wisely Norway

Led by GP and academic Dr Hjörleifsson, Choosing Wisely Norway is an initiative to encourage clinicians and patients to discuss when investigations and treatments are useful and when they are not. In Norway there is strong professional commitment to avoiding unnecessary testing, and the Norwegian College of General Practice adopted a position paper on overdiagnosis in 2016 – the first

of its kind to be adopted by a national medical college. I am working with Dr Hjörleifsson on a workshop for WONCA 2025 and will continue my collaboration with him throughout the fellowship. In the 2nd year of the PhD I will visit Dr Hjörleifsson and the GP academic community at the University of Bergen to learn more about their work advocating for prudent testing in primary care in Norway and consider how this could be translated to a UK context.

Lab Tests Online (UK), NHS Digital, and Pathology Tests Explained (Australia)

My supervisor Dr Watson works closely with Lab Tests Online, and my ACHEEV mentor Dr Shih is undertaking a project with Pathology Tests Explained (the Australian counterpart), on which I am a technical advisor. These organisations provide online resources to help patients understand their blood test results. Dr Watson has also been collaborating with NHS Digital on the Accessible Results study to provide guidance on NHS app design for online results access. I will work with these three organisations to feedback my research findings and provide recommendations of how online blood test resources could be improved.

Community collaborations

Social Action for Health (SAfH): East London community health charity.

SAfH have extensive experience running community health research projects and focus groups in collaboration with NHS and academic partners. I will work with SAfH to assemble a group of people to participate in the Forum Theatre workshops in WP4.

Improbable: Improvisation-based theatre company (https://www.improbable.co.uk/)

Alongside producing award-winning West End shows, Improbable has a research wing which uses improvisation to interpret research findings and bring them to new audiences. Improbable's Executive Director has agreed to collaborate on the project. Theatre-makers from Improbable will help to design the play and facilitate the Forum Theatre workshops.

Host organisation support statement (1000 words)

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